



DEALER BEST PRACTICE SERIES

Proper Gear Reminder for Trucks Ascending Grades

Application Maintenance Site Management Component Rebuild Safety MARC Management

Proper Gear Reminder for Trucks Ascending Grades1

1.0 Best Practice Description.....2

2.0 Implementation Steps2

3.0 Benefits.....2

4.0 Resources Required3

5.0 Supporting Attachments3

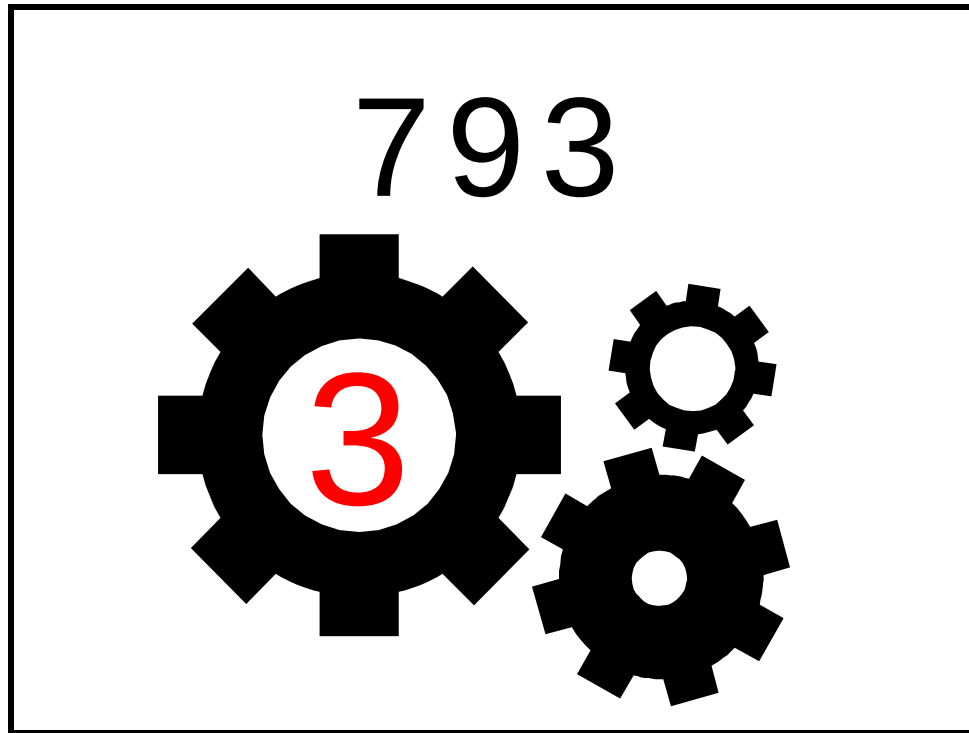
6.0 Related Best Practices3

7.0 Acknowledgements.....3

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Best Practice Description

This Best Practice deals with reminding truck operators to engage and lock the proper transmission gear for ascending grades with a loaded truck. This will allow the truck to ascend the grade without downshifting on the grade. To remind truck operators of this practice, mines have placed proper gear signs in locations prior to beginning of the uphill grades. The sign indicates to the operator which gear should be engaged and locked prior to ascending the grade.



A suggested proper gear sign for a 793:

2.0 Implementation Steps

- Establish the proper gear to be selected and engaged for each grade in the mine for each truck model. This can be accomplished by first ascending the grade with a loaded truck and allowing the transmission to downshift to the lowest gear it needs to ascend the grade. This lowest gear is the “proper” gear for ascending the grade.
- Train the operators.
- Place correct gear indicator signs prior to each of the uphill grades. Signs should be placed at distance great enough to allow the operator to slow, engage and lock the proper gear.

3.0 Benefits

Depending upon the grade and payload, there is a particular gear in which the truck will ascend the grade the fastest without downshifting. Benefits of eliminating unnecessary downshifts on the grade are:

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Correct Gear Reminder for Trucks Ascending Grades	DATE 21 SEP 2006	CHG NO 02	NUMBER 0906-1.2-1017

- 1) Preventing material spillage on the haul road that leads to tire cuts or other tire damage,
- 2) Extending power train component life by reducing the number of full power downshifts,
- 3) Preventing ruts from forming on the grade caused by the abrupt tire slippage occurring during a downshift.

4.0 Resources Required

All that is required are signs appropriately placed before uphill grades that inform the operator of the proper gear to use while ascending the grade. The sign may be as simple as a symbol of a gear with a number at the center. The number indicates the correct gear to be selected and engaged before ascending the grade. Such a pictograph avoids any language problems. In situations where different truck models require different gears to be selected and engaged, the sign may indicate the particular truck model: 793, 797, etc.

5.0 Supporting Attachments

None applicable.

6.0 Related Best Practices

None applicable

7.0 Acknowledgements

This Best Practice was authored by:

Fred Ringness
Caterpillar Global Mining
Product Support Manager – Site Capabilities
Ringness_Fred@cat.com
309-675-4882

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Correct Gear Reminder for Trucks Ascending Grades	DATE 21 SEP 2006	CHG NO 02	NUMBER 0906-1.2-1017
---	---------------------	-----------------	-------------------------